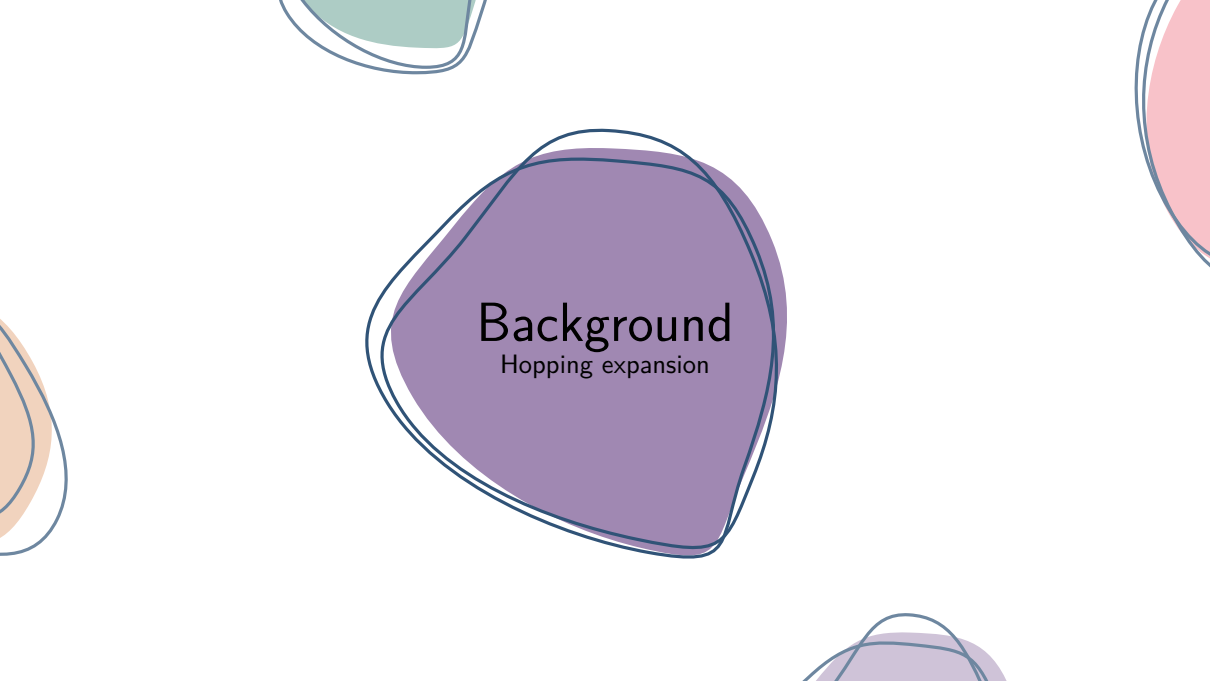


Neural-network approaches for preconditioning the Dirac equation in lattice QCD

Simon Pfahler

University of Regensburg

May 26, 2026

The background features several overlapping, hand-drawn organic shapes in muted colors: teal, orange, pink, and purple. The central focus is a large, irregular purple shape with a dark blue outline, which contains the text.

Background

Hopping expansion

Problem statement

Bigger picture

- ▶ Lattice QCD calculations are expensive
- ▶ One of the most expensive parts is solving the Dirac equation
- ▶ Preconditioners can accelerate this

Preconditioners

- ▶ State of the art: algebraic multigrid (AMG)
- ▶ But AMG needs costly setup
- ▶ Can this setup be avoided?
- ▶ **Goal:** Preconditioner competitive with AMG, but with very little (or even no) setup

Lattice Definitions 1

- ▶ We simulate QCD on a discrete space-time lattice
→ fermion fields $\psi \in \mathbb{C}^{V \times 4 \times 3}$
→ gauge fields $U \in SU(3)^{4 \times V}$
- ▶ Gauge transformations $\Omega \in SU(3)^V$ transform the fields as

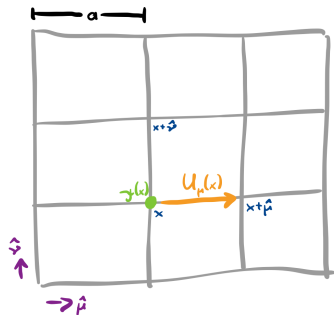
$$\psi(x) \rightarrow \Omega(x)\psi(x)$$

$$U_\mu(x) \rightarrow \Omega(x)U_\mu(x)\Omega^\dagger(x + \hat{\mu})$$

- ▶ This leads to parallel-transport operators

$$H_\mu\psi(x) = U_\mu^\dagger(x - \hat{\mu})\psi(x - \hat{\mu})$$

These transform as $H_\mu\psi(x) \rightarrow \Omega(x)H_\mu\psi(x)$



Lattice Definitions 2

Dirac equation: $D\psi = \phi$

We use the Wilson(-clover) Dirac operator

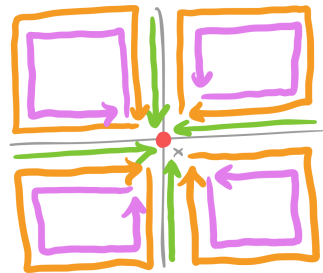
$$D_{WC} = D_W - \frac{c_{sw}}{4} \sum_{\mu, \nu} \sigma_{\mu\nu} F_{\mu\nu}$$

$$D_W = (m + 4) - \sum_{\mu} \left(\frac{1 - \gamma_{\mu}}{2} H_{-\mu} + \frac{1 + \gamma_{\mu}}{2} H_{+\mu} \right)$$

$$F_{\mu\nu} = \frac{1}{8} (Q_{\mu\nu} - Q_{\nu\mu})$$

$$Q_{\mu\nu} = P_{+\nu, +\mu} + P_{-\mu, +\nu} + P_{-\nu, -\mu} + P_{+\mu, -\nu}$$

$$P_{\pm\mu, \pm\nu} = H_{\mp\nu} H_{\mp\mu} H_{\pm\nu} H_{\pm\mu}$$



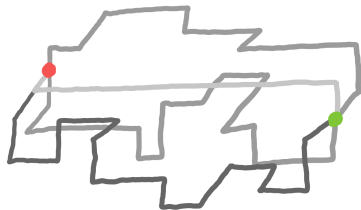
action of D_{WC} on $\psi(x)$

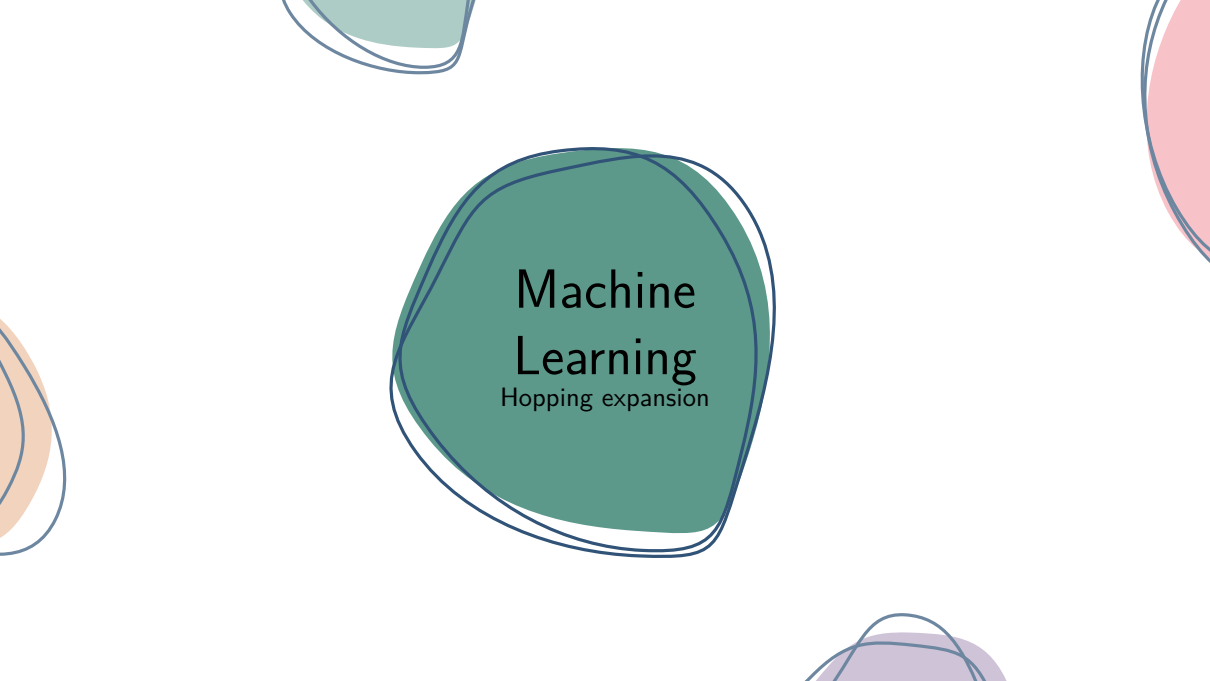
Hopping expansion

- ▶ We can write $D_W = (m + 4)(\mathbb{1} - \kappa H)$ with hopping parameter $\kappa = \frac{1}{2(m+4)}$
- ▶ For $\kappa < \frac{1}{8}$, the inverse of D_W is given by

$$D_W^{-1} = \frac{1}{m+4}(\mathbb{1} - \kappa H)^{-1} = \frac{1}{m+4} \sum_{n=0}^{\infty} \kappa^n H^n$$

- ▶ Can we learn systematic hopping-style expansions using Machine Learning?
 $\Rightarrow$ Yes, let's see how!



The background is white with several large, hand-drawn, overlapping circles in various colors: teal, orange, pink, and purple. The central teal circle contains the text.

Machine Learning

Hopping expansion

Network Architecture



Dense linear (L) layer

- ▶ n spin-color fields
 $\varphi \in \mathbb{C}^{L_x \times L_y \times L_z \times L_t \times N_c \times N_s}$
- ▶ build m linear combinations of the input fields, with spin matrices $W_{ij} \in \mathbb{C}^{N_s \times N_s}$ as weights:

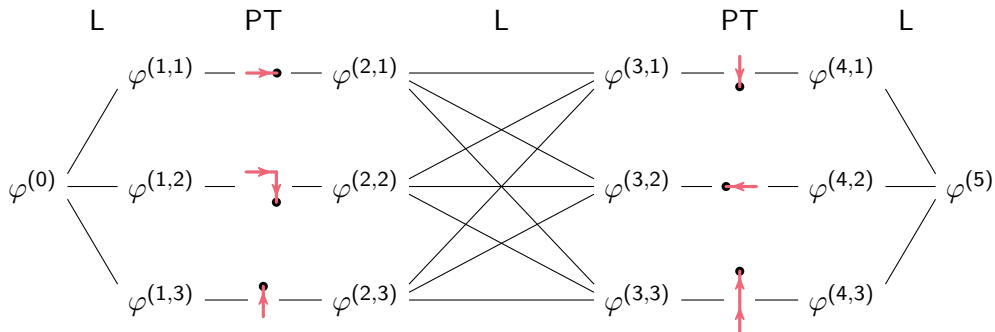
$$\psi_i(x) \stackrel{\text{L}}{=} \sum_{j=1}^n W_{ij} \varphi_j(x)$$

Parallel Transport (PT) layer

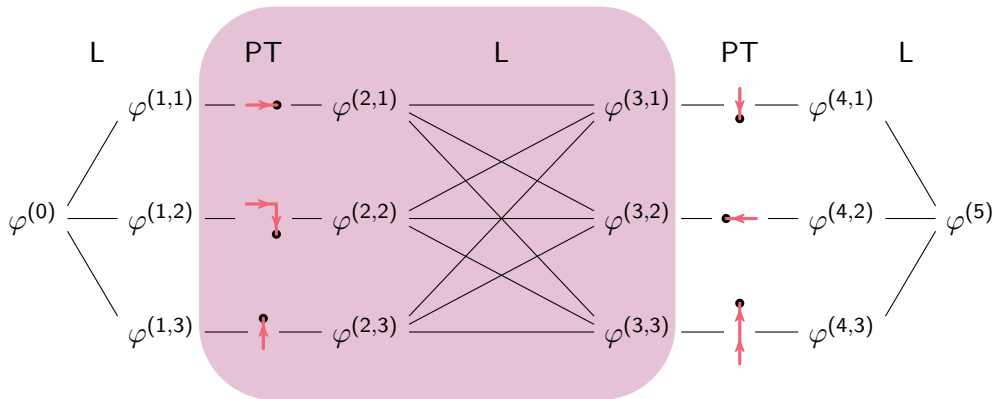
- ▶ n spin-color fields
 $\varphi \in \mathbb{C}^{L_x \times L_y \times L_z \times L_t \times N_c \times N_s}$
- ▶ apply a different parallel transport $p^{(i)}$ to every input field:

$$\psi_i(x) \stackrel{\text{PT}}{=} T_{p^{(i)}} \varphi_i(x)$$

Network Architecture



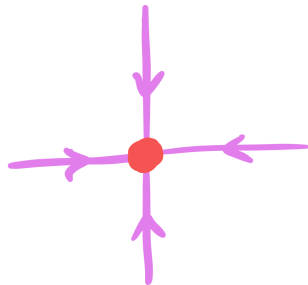
Network Architecture



Choice of PT paths

The most natural choice of PT paths is
only nearest-neighbor hops:

$$P = \{1\} \cup \{H_p | p \in \{\pm 1, \pm 2, \pm 3, \pm 4\}\}$$



Score function



- ▶ To train our network, we need a score function
- ▶ Ideally, we want to optimize the condition number of MD ,

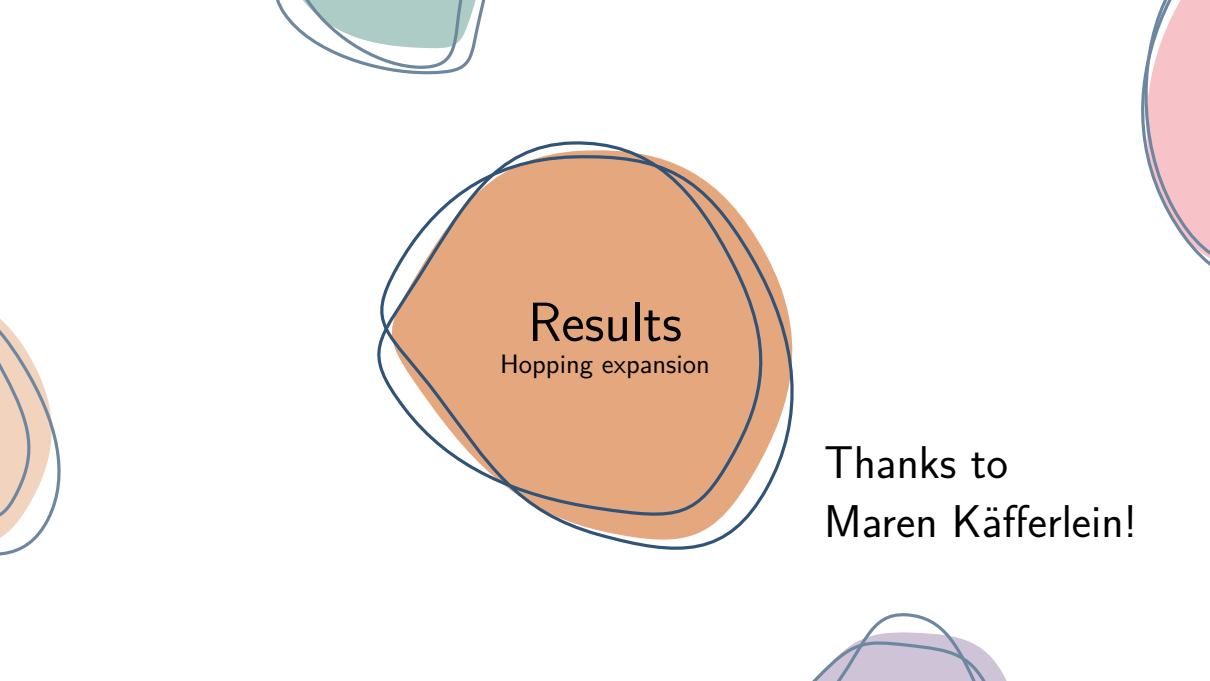
$$\kappa(MD) = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

→ Expensive and unstable ☹

- ▶ Instead, we use the surrogate cost function

$$C = \|MD\varphi - \varphi\|^2$$

with random fermion fields φ

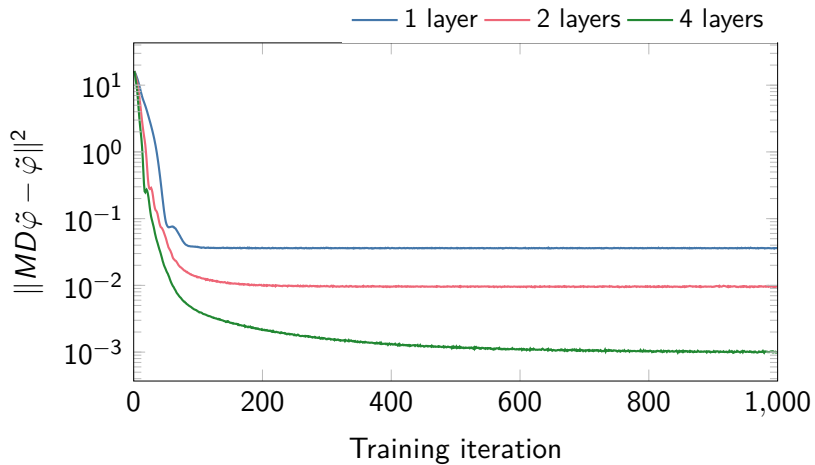
The background features several abstract, hand-drawn shapes in various colors: a teal shape at the top left, a pink shape at the top right, an orange shape on the left, and a purple shape at the bottom. The central focus is a large orange shape with a dark blue outline, containing the text.

Results

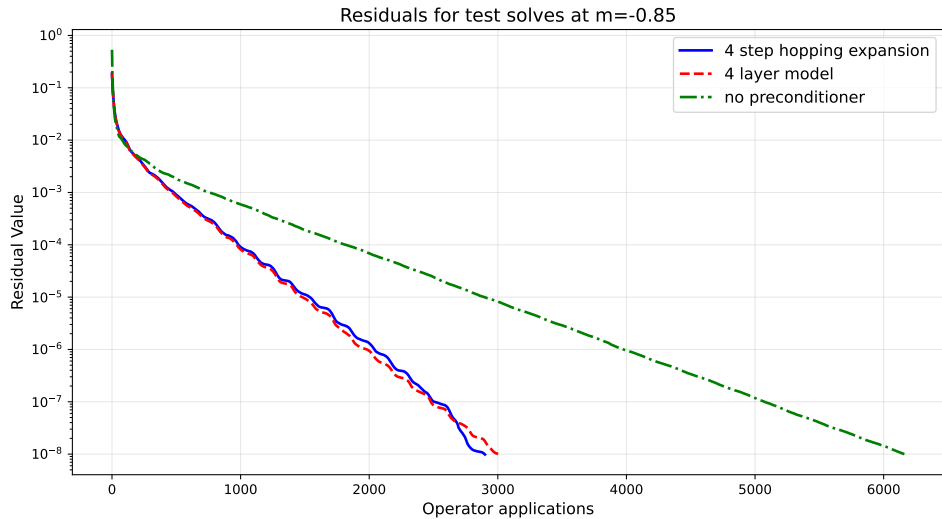
Hopping expansion

Thanks to
Maren Käfferlein!

Training



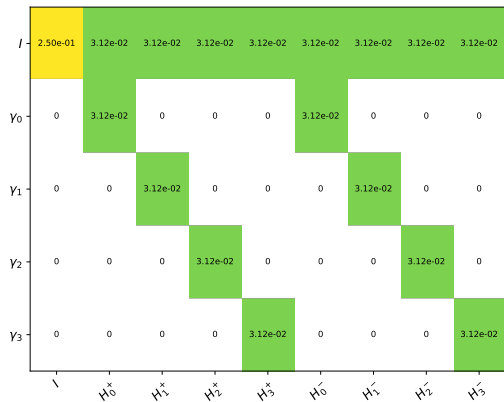
Test solves



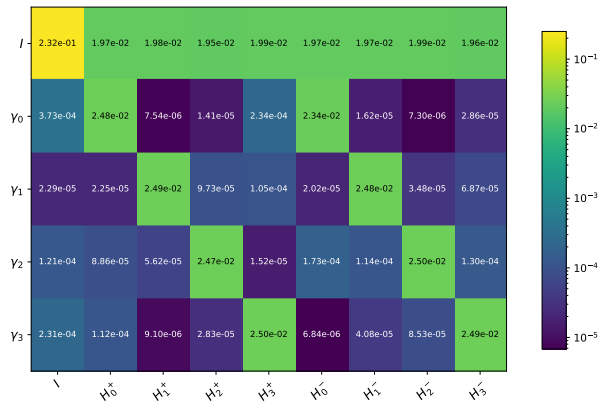
Analytical structure

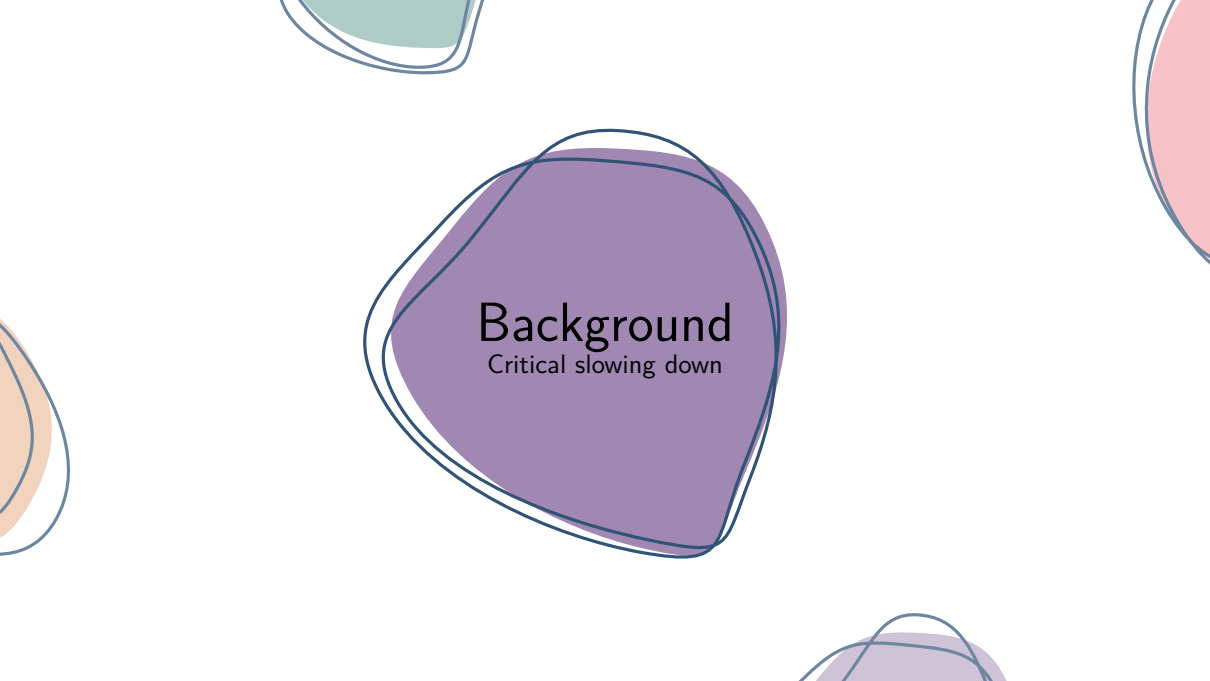
1 layer

Hopping expansion



Learned coefficients



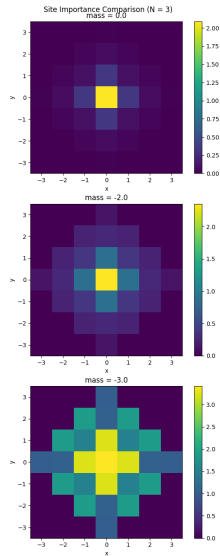
The background features several abstract, organic shapes in muted colors: a teal shape at the top left, a pink shape at the top right, an orange shape on the left edge, and a purple shape at the bottom right. The central focus is a large, irregular purple shape with a dark blue outline, which contains the text.

Background

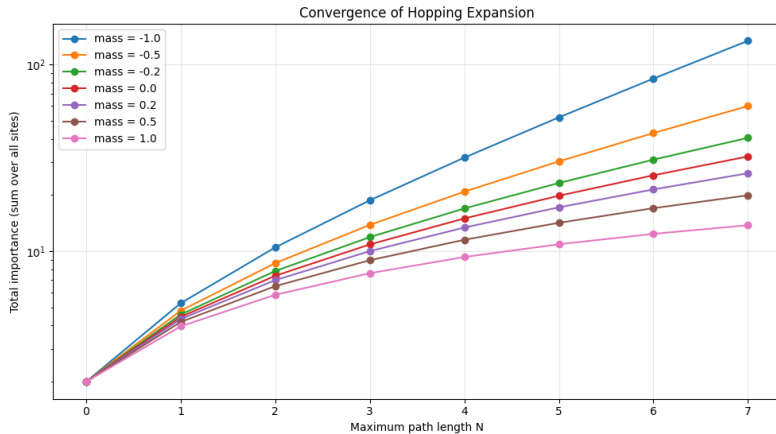
Critical slowing down

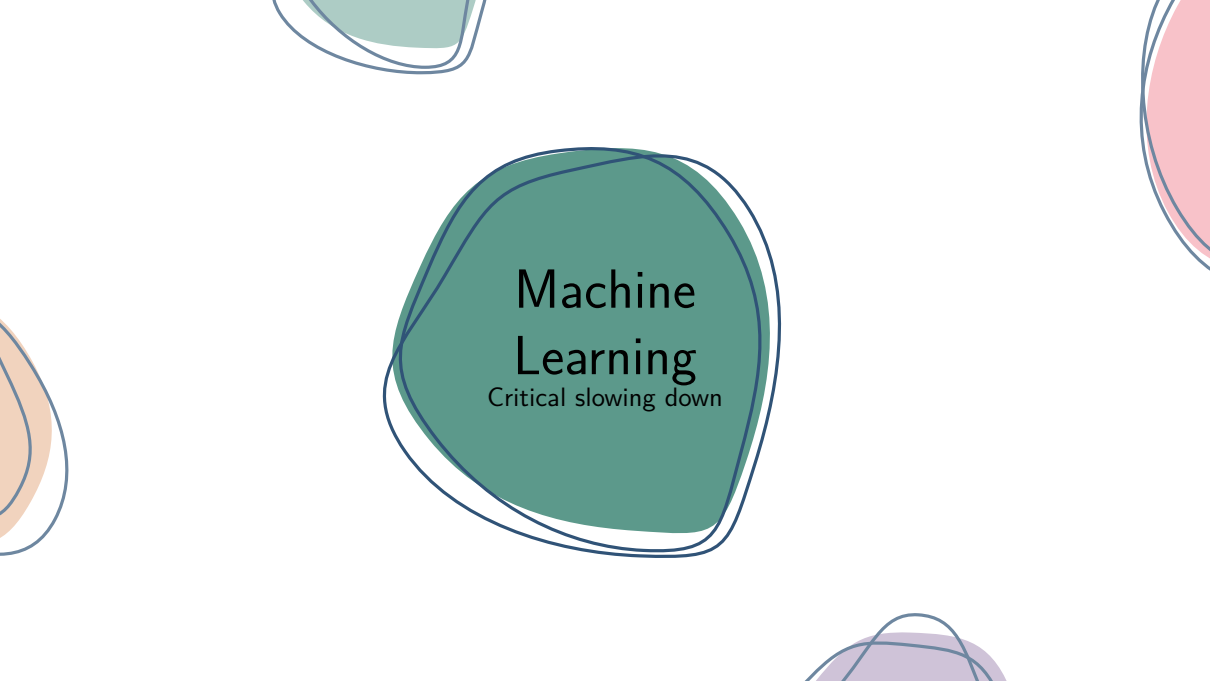
Critical slowing down

- ▶ For $\kappa = \frac{1}{2(m+4)} < \frac{1}{8}$, the hopping expansion does not converge anymore, as long paths dominate
- ▶ Near $\kappa \approx \frac{1}{8}$, we need an increasing amount of layers to build a good approximation of D^{-1}
- ▶ How can we make our networks more efficient in this regime?



Critical slowing down

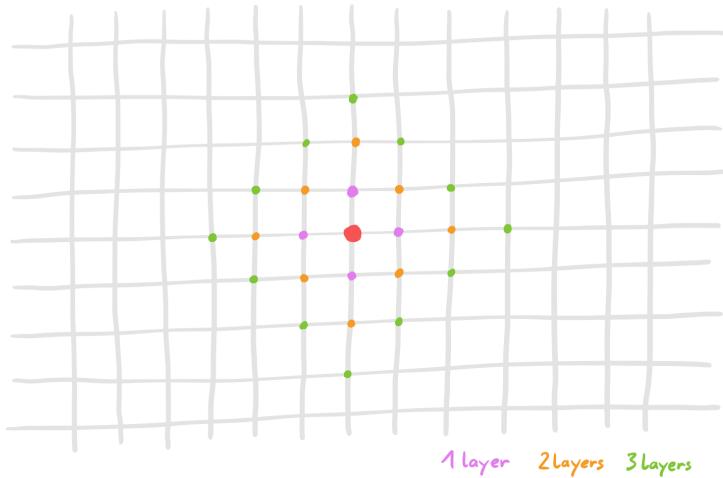


The background is white with several large, overlapping, hand-drawn style shapes in muted colors: teal, orange, pink, and purple. The central teal shape is the largest and contains the text.

Machine Learning

Critical slowing down

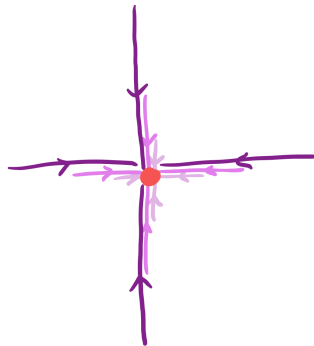
PT Paths 1



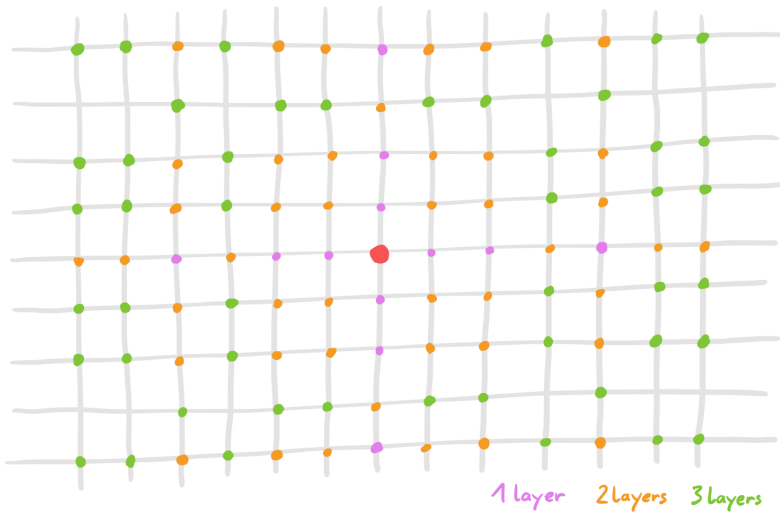
PT Paths 2

To allow for longer paths without increasing the number of paths too much, we choose

$$P = \{1\} \cup \{H_p^n | p \in \{\pm 1, \pm 2, \pm 3, \pm 4\}, \\ n \in \{2^0, 2^2, \dots, L_p/2\}\}$$



PT Paths 3



Score function

► Standard choice:

$$C = \|MD\varphi - \varphi\|^2$$

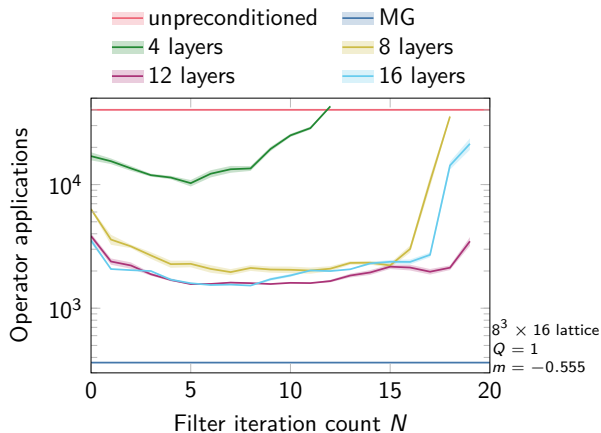
with random field φ

► Filtered cost function:

$$C = \|MD\tilde{\varphi} - \tilde{\varphi}\|^2$$

where $\tilde{\varphi}$ is obtained through N iterations of GMRES for $D\varphi = 0$ with random initial guess

⇒ Stronger emphasis on the low modes!



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Results

Critical slowing down

Gauge configurations

$8^3 \times 16$ ensemble¹:

- ▶ Wilson action
- ▶ quenched, heat bath
- ▶ Wilson-clover Dirac operator
- ▶ configurations with topological charge $|Q| \in \{0, 1, 2\}$

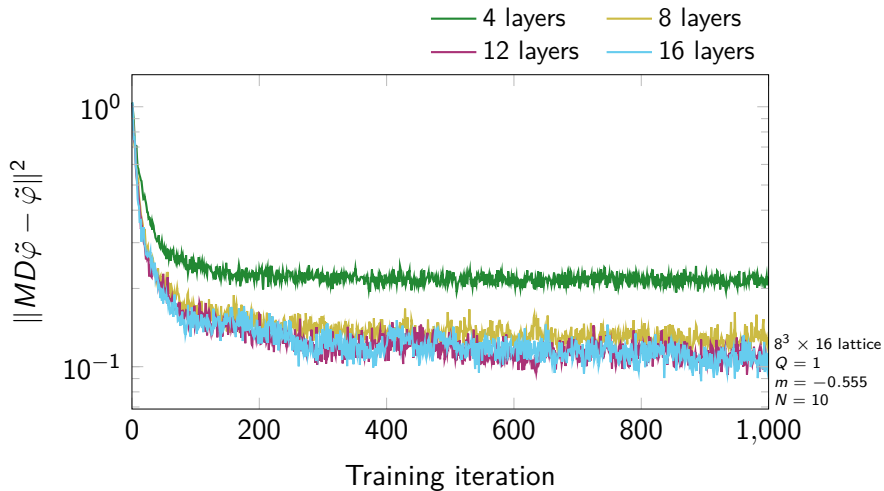
$16^3 \times 32$ ensemble²:

- ▶ Wilson action
- ▶ quenched, HMC
- ▶ Wilson-clover Dirac operator
- ▶ configurations with topological charge $|Q| \in \{0, 4\}$

¹ doi.org/10.5281/zenodo.15018324

² doi.org/10.5281/zenodo.17494829

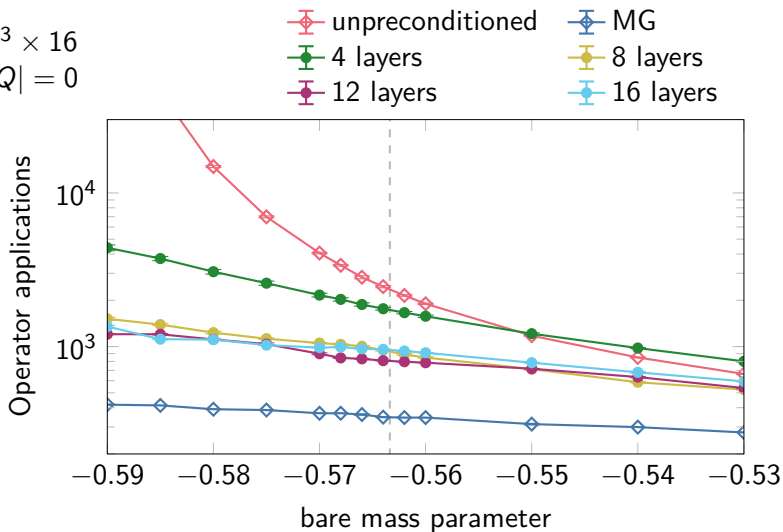
Training cost



Critical slowing down

$8^3 \times 16$

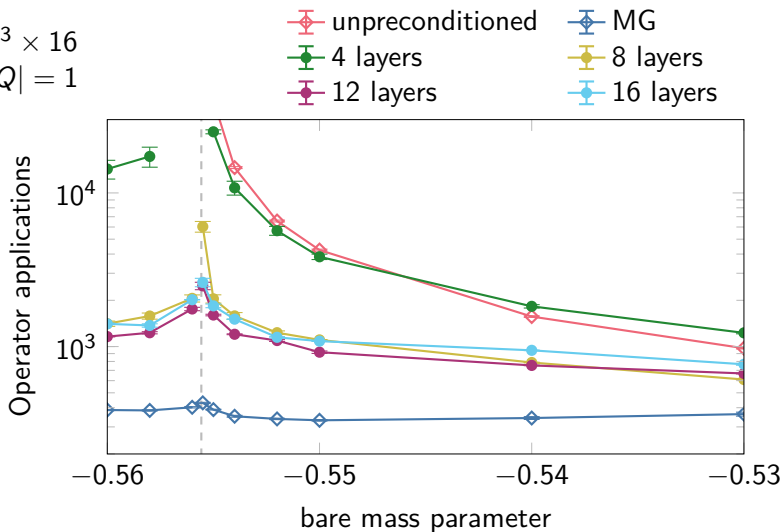
$|Q| = 0$



Critical slowing down

$$8^3 \times 16$$

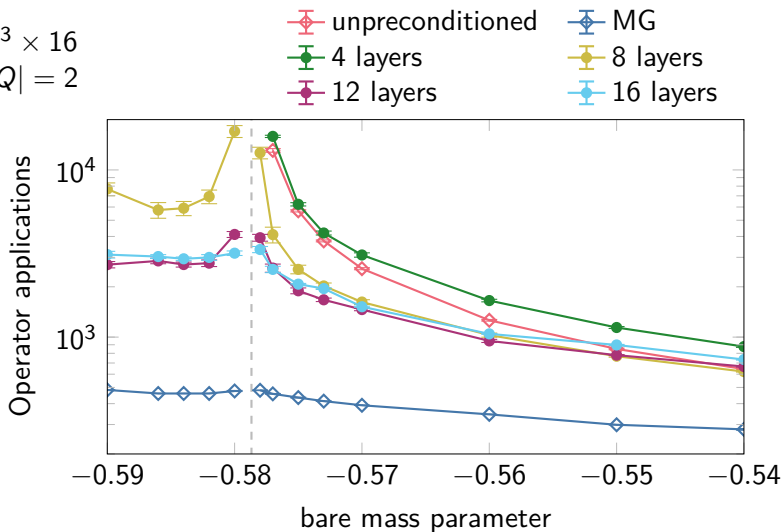
$$|Q| = 1$$



Critical slowing down

$$8^3 \times 16$$

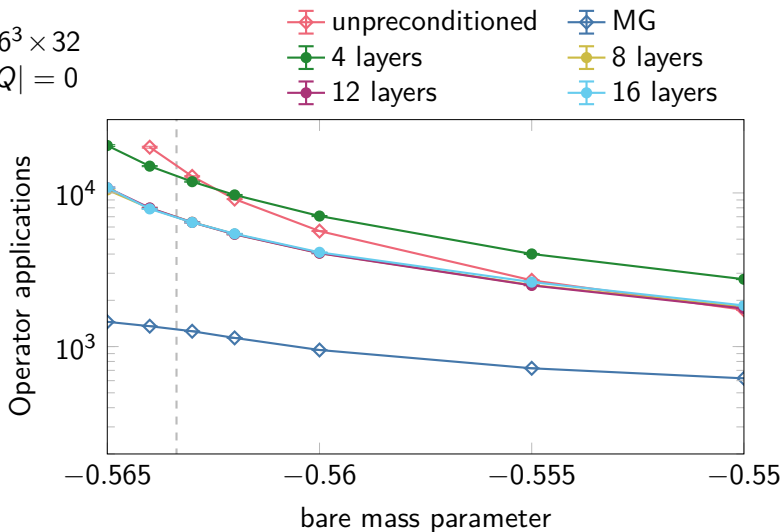
$$|Q| = 2$$



Critical slowing down

$16^3 \times 32$

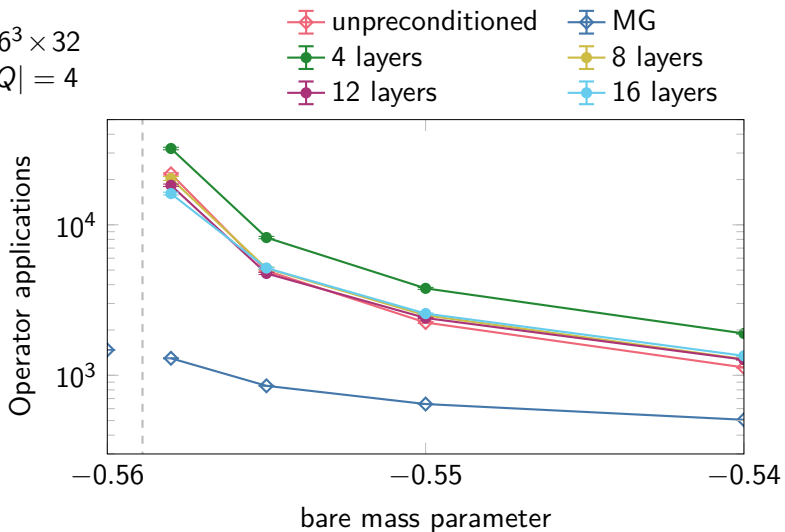
$|Q| = 0$



Critical slowing down

$16^3 \times 32$

$|Q| = 4$



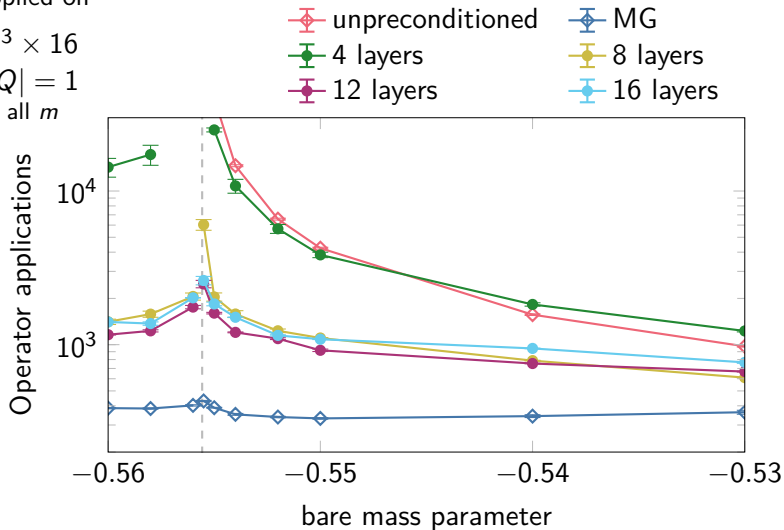
Transfer

trained and
applied on

$$8^3 \times 16$$

$$|Q| = 1$$

all m



Transfer

trained on

$$8^3 \times 16$$

$$|Q| = 0$$

$$m = -0.56$$

applied on

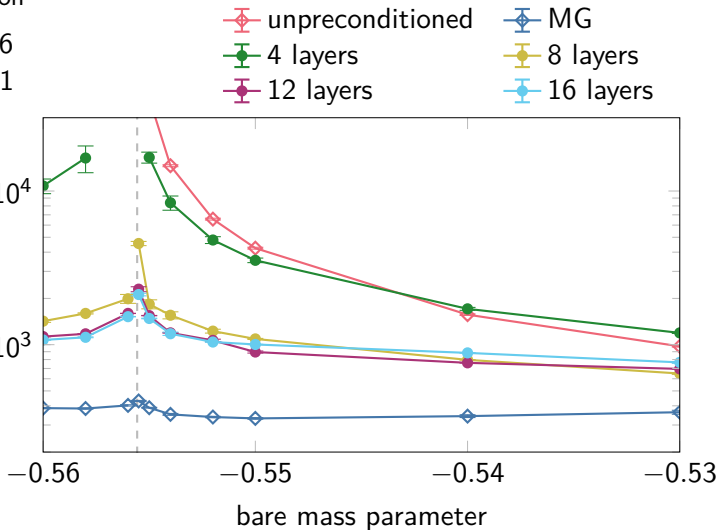
$$8^3 \times 16$$

$$|Q| = 1$$

$$\text{all } m$$

$\rightarrow$

Operator applications



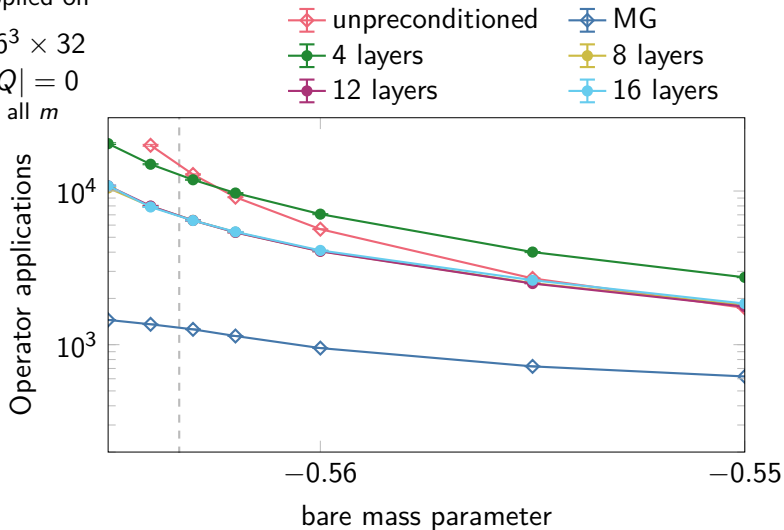
Transfer

trained and
applied on

$$16^3 \times 32$$

$$|Q| = 0$$

all m



Transfer

trained on

$$8^3 \times 16$$

$$|Q| = 0$$

$$m = -0.56$$

applied on

$$16^3 \times 32$$

$$|Q| = 0$$

$$\text{all } m$$

→

unpreconditioned

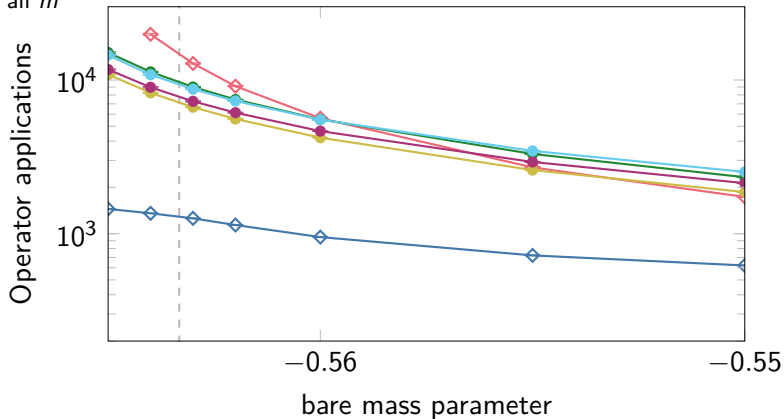
4 layers

12 layers

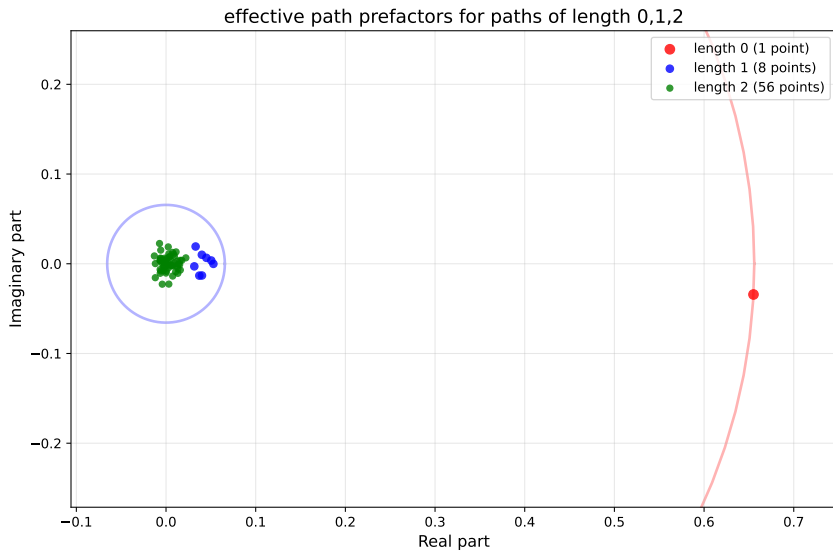
MG

8 layers

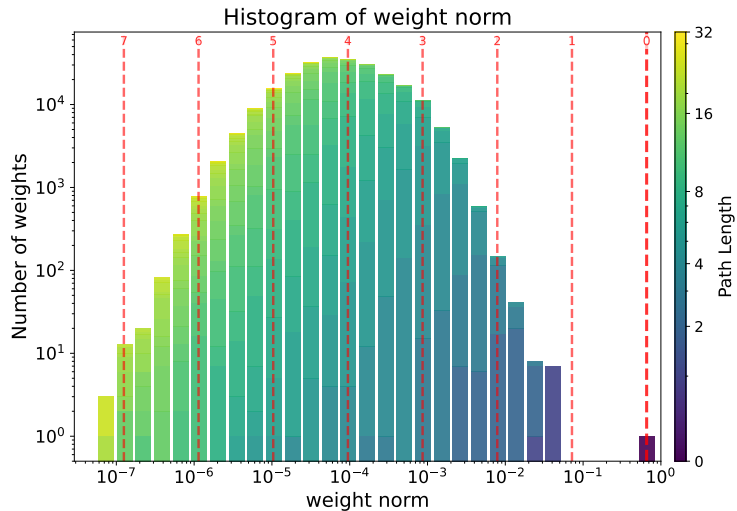
16 layers



Weights analysis

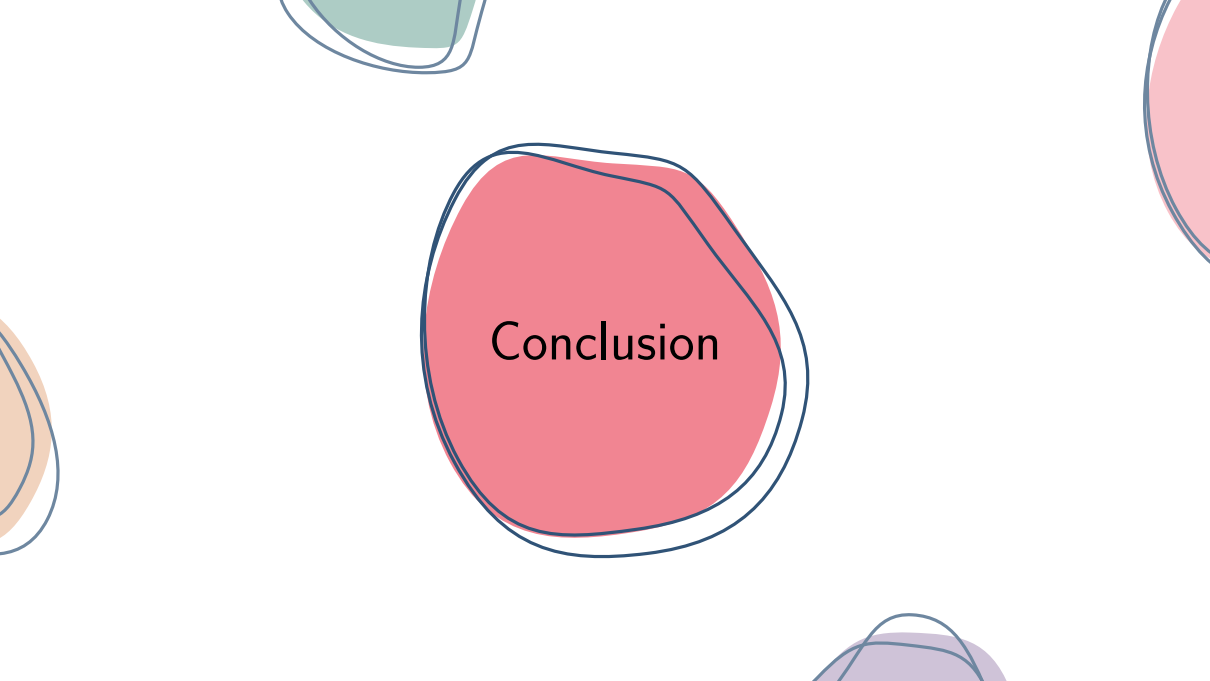


Weights analysis



Weights look systematic,
but not like hopping
expansion!

→ new analytic formula
for D_{WC}^{-1} ?



Conclusion

Conclusion

Goals

- ☐ Build a preconditioner using ML that mitigates critical slowing down
- ☐ Learn a general analytic formula
→ No costly setup/trivial transfer
- ☐ Handle topological modes nicely



Slides and code:
simon-pfahler.github.io

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Goals

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Next steps

- ▶ Extract the analytic expansion the models learned
- ▶ Understand why the architecture fails for topological modes
- ▶ Explore models to generate representative near-null space vectors economically

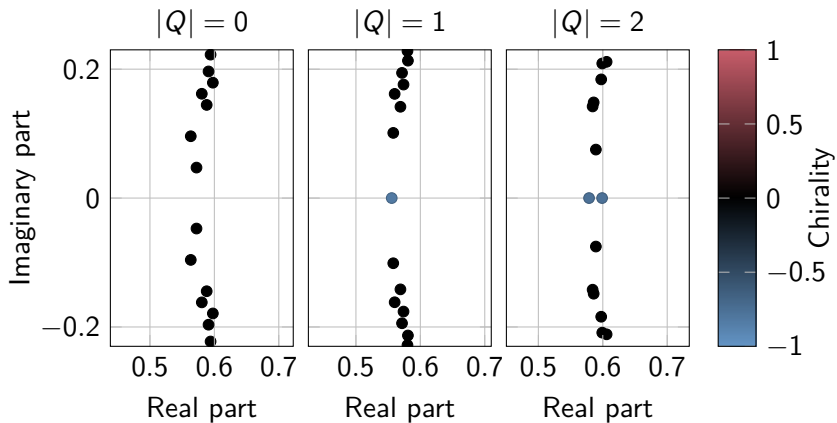


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Bonus
Slides

Spectra

$8^3 \times 16$ lattice



Spectra

$16^3 \times 32$ lattice

